MXB103 Cheat Sheet Side 1

Calculator in **RADIANS**; log = ln. Write the general formula first (it usually earns a mark).

W1 Floating point

$F(\beta,k,m,M)$: $\pm d_1.d_2\cdots d_k \times \beta^e$, $d_1 \neq 0$, $m \le e \le M$ (k counts *all* digits)

- $S=\beta^m \quad L=(\beta-\beta^{1-k})\beta^M \quad u=\frac{1}{2}\beta^{1-k}$
- abs $=|x-fl(x)|$, rel $=\dfrac{|x-fl(x)|}{|x|}\le u$ (divide by the **true** value)
- dec→bin: integer ÷2, read remainders **up**; fraction ×2, read integer parts **down**
- $|x|<S$: underflow → 0; $|x|>L$: overflow → ∞

How to round x to $fl(x)$:

1. write x in base β, normalise to $d_1.d_2\cdots\times\beta^e$
2. cut after digit k: $d_1.d_2\cdots d_k \,|\, d_{k+1}\cdots$
3. look at the **first dropped digit** d_{k+1}: binary: 1 ⇒ round up, 0 ⇒ chop; decimal: ≥ 5 ⇒ round up, ≤ 4 ⇒ chop
4. round up = add 1 to the last kept digit (carry like normal addition)
5. if it carries to the front ($1.111|1 \to 10.000$), renormalise: $1.000\times\beta^{e+1}$
6. check $m\le e\le M$ **after** rounding

e.g. $F(2,5,-4,4)$: $S=2^{-4}=0.0625$, $L=(2-2^{-4})2^4=31$, $u=\frac{1}{2}2^{-4}=0.03125$
0.1: ×2 gives digits 0 0 0 1 1 0 0 1 1 …
$0.1=1.1001\,|\,1001\ldots\times 2^{-4}$
dropped digit 1 ⇒ round up: $1.1010\times 2^{-4}=0.1015625$
rel err $=\frac{0.0015625}{0.1}=0.0156\le u$ ✓
$31.5=11111.1_2=1.1111\,|\,1\times 2^4$
round up: $10.0000\times 2^4=1.0000\times 2^5$
$e=5>M=4$ ⇒ **overflow**
$0.05<S=0.0625$ ⇒ **underflow** to 0

W2 Cancellation & Taylor

Cancellation: subtracting nearly-equal *rounded* numbers loses digits. Fix: rewrite so there is no subtraction.

- $\frac{-b+\sqrt{b^2-4ac}}{2a}=\frac{-2c}{b+\sqrt{b^2-4ac}}$ $(b>0)$
- $1-\cos x=\frac{\sin^2 x}{1+\cos x}$; $\sqrt{x+1}-\sqrt{x}=\frac{1}{\sqrt{x+1}+\sqrt{x}}$
- $\log(x+1)-\log x=\log(1+\frac{1}{x})$

Taylor polynomial (degree n, about x_0):

$$P_n(x)=\sum_{k=0}^{n}\frac{f^{(k)}(x_0)}{k!}(x-x_0)^k$$

$$R_n(x)=f(x)-P_n(x)=\frac{f^{(n+1)}(c)}{(n+1)!}(x-x_0)^{n+1}$$

(c between x_0 and x) **Bound on an interval:** degree n ⇒ use derivative $n+1$

$$|R_n(x)|\le\frac{\max|f^{(n+1)}(c)|}{(n+1)!}\cdot\max|x-x_0|^{n+1}$$

Take **both** maxes over the interval (check the interior too).

e.g. $f=\sin x$, $x_0=\frac{\pi}{6}$, $x\in[0,\frac{\pi}{3}]$
(a) degree 1 ($n=1$):
$P_1(x)=\sin\frac{\pi}{6}+\cos\frac{\pi}{6}\,(x-\frac{\pi}{6})=\frac{1}{2}+\frac{\sqrt{3}}{2}(x-\frac{\pi}{6})$
(b) $n+1=2$, so use $f''(x)=-\sin x$:
$R_1(x)=\dfrac{-\sin c}{2!}(x-\frac{\pi}{6})^2$, c between $\frac{\pi}{6}$ and x
$\max|\sin c|$ on $[0,\frac{\pi}{3}]=\sin\frac{\pi}{3}=\frac{\sqrt{3}}{2}$
$\max|x-\frac{\pi}{6}|=\frac{\pi}{6}$ (distance from x_0, not from 0)
$|R_1|\le\dfrac{\sqrt{3}/2}{2}\left(\dfrac{\pi}{6}\right)^2=\dfrac{\sqrt{3}\,\pi^2}{144}\approx 0.1187$

W3–4 ODEs $y'=f(t,y)$, $y(a)=\alpha$

$h=\frac{b-a}{n}$, $t_i=a+ih$, $w_i\approx y_i=y(t_i)$. Output: a *table* of (t_i,w_i).

- **Euler:** $w_{i+1}=w_i+hf(t_i,w_i)$
- **Taylor 2:** $w_{i+1}=w_i+hf(t_i,w_i)+\frac{h^2}{2}f'(t_i,w_i)$
  ! $f'(t,y)=f_t(t,y)+f_y(t,y)\cdot f(t,y)$ (chain rule)
- **Mod. Euler:** $k_1=hf(t_i,w_i)$, $k_2=hf(t_i+h,w_i+k_1)$
  $w_{i+1}=w_i+\frac{1}{2}(k_1+k_2)$
- **RK4:** $k_1=hf(t_i,w_i)$, $k_2=hf(t_i+\frac{h}{2},w_i+\frac{k_1}{2})$
  $k_3=hf(t_i+\frac{h}{2},w_i+\frac{k_2}{2})$, $k_4=hf(t_i+h,w_i+k_3)$
  $w_{i+1}=w_i+\frac{1}{6}(k_1+2k_2+2k_3+k_4)$

e.g. $y'=2t-y^2$, $y(0)=1$, $h=0.1$
Euler: $w_1=1+0.1\,f(0,1)=1+0.1(0-1)=0.9$
$w_2=0.9+0.1\,f(0.1,0.9)=0.9+0.1(0.2-0.81)=0.839$
Taylor 2: $f_t=2$, $f_y=-2y$
$f'(t,y)=2-2y(2t-y^2)$, $f'(0,1)=2-2(1)(-1)=4$
$w_1=1+0.1(-1)+\frac{0.01}{2}(4)=0.92$
Mod. Euler: $k_1=0.1\,f(0,1)=-0.1$
$k_2=0.1\,f(0.1,0.9)=-0.061$
$w_1=1+\frac{1}{2}(-0.1-0.061)=0.9195$
RK4: $k_1=-0.1$, $k_2=0.1\,f(0.05,0.95)=-0.08025$
$k_3=0.1\,f(0.05,0.959875)=-0.082136$
$k_4=0.1\,f(0.1,0.917864)=-0.064247$
$w_1=1+\frac{1}{6}(-0.1-0.1605-0.164272-0.064247)=0.918497$

	Euler	Taylor 2	Mod. Euler	RK4
local	$O(h^2)$	$O(h^3)$	$O(h^3)$	$O(h^5)$
global	$O(h)$	$O(h^2)$	$O(h^2)$	$O(h^4)$
needs f'?	no	**yes**	no	no

Local $O(h^{p+1})\times\frac{b-a}{h}$ steps = global $O(h^p)$. Halve h ⇒ error ÷2^p. e.g. err 0.004 → 0.001 when h halves: $4=2^p$, $p=2$.
Accuracy: "(b) is more accurate: **second order** $O(h^2)$ vs Euler's **first order** $O(h)$."
Derive Taylor 2:

1. Taylor-expand $y(t)$ about t_i:
   $y(t)=y(t_i)+y'(t_i)(t-t_i)+\frac{y''(t_i)}{2}(t-t_i)^2+\frac{y'''(c)}{6}(t-t_i)^3$
2. put $t=t_{i+1}$, so $t_{i+1}-t_i=h$:
   $y_{i+1}=y_i+hy'(t_i)+\frac{h^2}{2}y''(t_i)+O(h^3)$
3. use the ODE: $y'(t_i)=f(t_i,y_i)$, $y''(t_i)=f'(t_i,y_i)$
4. drop $O(h^3)$, replace $y\to w$:
   $w_{i+1}=w_i+hf(t_i,w_i)+\frac{h^2}{2}f'(t_i,w_i)$

Derive Mod. Euler: steps 1–3 as above, then

4. forward difference for f':
   $f'(t_i,y_i)=\dfrac{f(t_{i+1},y_{i+1})-f(t_i,y_i)}{h}+O(h)$
5. substitute into step 2 (the $\frac{h^2}{2}\cdot O(h)$ is $O(h^3)$):
   $y_{i+1}=y_i+\frac{h}{2}\left[f(t_i,y_i)+f(t_{i+1},y_{i+1})\right]+O(h^3)$
6. y_{i+1} is on both sides, so inside f use Euler:
   $y_{i+1}\approx w_i+hf(t_i,w_i)$
7. result: $w_{i+1}=w_i+\frac{h}{2}\left[f(t_i,w_i)+f(t_{i+1},w_i+hf(t_i,w_i))\right]$

MATLAB: `t`, `w` start at a,α; `h=(b-a)/n`; `f(t,w)` not `f(w,t)`; `fdash` has `*f(t,y)`; RK4 $\frac{k}{2}$, full k_3, `/6`; `t=t+h` after the update.

W5 Lagrange

$n+1$ distinct points ⇒ a **unique** P_n of degree $\le n$ (Lagrange, Newton DD and Newton FD all give the same polynomial).

$$P_n(x)=\sum_{i=0}^{n}L_{n,i}(x)\,y_i,\quad L_{n,i}(x)=\prod_{j\neq i}\frac{x-x_j}{x_i-x_j}$$

- $L_{n,i}(x_i)=1$, $L_{n,i}(x_j)=0$; **check** $\sum_i L_{n,i}(x)=1$
- error: $f(x)-P_n(x)=\dfrac{f^{(n+1)}(c)}{(n+1)!}(x-x_0)(x-x_1)\cdots(x-x_n)$

e.g. $(1,1),(2,1.414),(3,1.732)$ from $f=\sqrt{x}$; estimate $\sqrt{1.5}$
$L_{2,0}(1.5)=\frac{(1.5-2)(1.5-3)}{(1-2)(1-3)}=\frac{0.75}{2}=0.375$
$L_{2,1}(1.5)=\frac{(1.5-1)(1.5-3)}{(2-1)(2-3)}=\frac{-0.75}{-1}=0.75$
$L_{2,2}(1.5)=\frac{(1.5-1)(1.5-2)}{(3-1)(3-2)}=\frac{-0.25}{2}=-0.125$
check: $0.375+0.75-0.125=1$ ✓
$P_2(1.5)=0.375(1)+0.75(1.414)-0.125(1.732)=1.219$
bound: $f'''(x)=\frac{3}{8}x^{-5/2}$, max on $[1,3]$ at $x=1$: $\frac{3}{8}$
$|f(1.5)-P_2(1.5)|\le\dfrac{3/8}{3!}|(0.5)(-0.5)(-1.5)|=0.0234$

MATLAB: `.*` on vectors; denominator uses (x_i-x_j), $j\neq i$; rel err `abs(P-f(a))/abs(f(a))`; `x(1)` is x_0.

W6 Divided differences

$f[x_i]=y_i$, and (divide by last x − first x of the span):

$$f[x_i..x_{i+k}]=\frac{f[x_{i+1}..x_{i+k}]-f[x_i..x_{i+k-1}]}{x_{i+k}-x_i}$$

$$P_n(x)=f[x_0]+\sum_{k=1}^{n}f[x_0,..,x_k]\prod_{j=0}^{k-1}(x-x_j)$$
$$=f[x_0]+f[x_0,x_1](x-x_0)+f[x_0,x_1,x_2](x-x_0)(x-x_1)+\cdots$$

Coefficients = the **top diagonal**. Adding a point = 1 more row + 1 more term.

e.g. $(0,1),(1,3),(3,2)$, estimate $f(2)$

x	$f[\cdot]$	$f[\cdot,\cdot]$	$f[\cdot,\cdot,\cdot]$
0	**1**	$\frac{3-1}{1-0}=$ **2**	$\frac{-0.5-2}{3-0}=$ **−0.8333**
1	3	$\frac{2-3}{3-1}=-0.5$	
3	2		

$P_2(x)=1+2(x-0)-0.8333(x-0)(x-1)$
$P_2(2)=1+4-1.6667=3.3333$

Expand a 3rd-order DD (1 mark/line):

$$f[x_0..x_3]=\frac{f[x_1,x_2,x_3]-f[x_0,x_1,x_2]}{x_3-x_0}$$
$$f[x_1,x_2,x_3]=\frac{f[x_2,x_3]-f[x_1,x_2]}{x_3-x_1}$$
$$f[x_0,x_1,x_2]=\frac{f[x_1,x_2]-f[x_0,x_1]}{x_2-x_0}$$
$$f[x_i,x_{i+1}]=\frac{y_{i+1}-y_i}{x_{i+1}-x_i}$$

then substitute everything back into line 1.

W7 Forward differences (equal h)

$h=x_1-x_0$ (all gaps equal), $s=\dfrac{x-x_0}{h}$
$\Delta y_i=y_{i+1}-y_i$, $\Delta^k y_i=\Delta^{k-1}y_{i+1}-\Delta^{k-1}y_i$

$$P_n(x)=\sum_{k=0}^{n}\binom{s}{k}\Delta^k y_0$$

where $\binom{s}{k}=\dfrac{s(s-1)\cdots(s-k+1)}{k!}$, $\binom{s}{0}=1$

$$P_n=y_0+s\Delta y_0+\frac{s(s-1)}{2!}\Delta^2 y_0+\frac{s(s-1)(s-2)}{3!}\Delta^3 y_0+\cdots$$

$\Delta^2 y_0=y_2-2y_1+y_0$, $\Delta^3 y_0=y_3-3y_2+3y_1-y_0$
Link: $f[x_0,..,x_k]=\dfrac{\Delta^k y_0}{k!\,h^k}$. Reliable inside $[x_0,x_n]$; extrapolation ($s<0$ or $s>n$) is not.

e.g. same data: $(1,1),(2,1.414),(3,1.732)$, estimate $\sqrt{1.5}$

x	y	Δy	$\Delta^2 y$
1	**1**	**0.414**	**−0.096**
2	1.414	0.318	
3	1.732		

Δy: $1.414-1$, $1.732-1.414$; $\Delta^2 y$: $0.318-0.414$
coefficients = the **top row**
$h=2-1=1$, $s=\frac{x-1}{1}=x-1$
$P_2=1+0.414s+\frac{s(s-1)}{2}(-0.096)$
$x=1.5\Rightarrow s=0.5$:
$P_2=1+0.207+\frac{(0.5)(-0.5)}{2}(-0.096)=1.219$
(same as W5)

MXB103 Cheat Sheet Side 2

W8 Bisection & fixed point

Bracket (2 marks): "$f(a) = \cdots < 0$, $f(b) = \cdots > 0$: opposite signs and f continuous, so by the IVT $[a,b]$ contains a root."

- **Bisection:** $p = \frac{a+b}{2}$; if $f(p)$ has the sign of $f(a)$ then $a \leftarrow p$, else $b \leftarrow p$. Answer = the latest midpoint; $|p_n - r| \le \frac{b-a}{2^n}$
- **Fixed point:** rewrite as $x = g(x)$, iterate $x_{n+1} = g(x_n)$. Converges if $g : [a,b] \to [a,b]$ and $|g'(x)| \le k < 1$ (sufficient, not necessary). Linear, rate $\approx |g'(p)|$

e.g. $f(x) = x^2 - 2$ on $[1,2]$: $f(1) = -1 < 0$, $f(2) = 2 > 0$

n	a	b	p	$f(p)$
1	1	2	1.5	$+0.25 \Rightarrow b = p$
2	1	1.5	1.25	$-0.4375 \Rightarrow a = p$
3	1.25	1.5	1.375	$-0.1094 \Rightarrow a = p$

answer after 3 iterations: 1.375, error $\le \frac{1}{2^3} = 0.125$

fixed point $x = \cos x$, $x_0 = 1$:
$x_1 = \cos 1 = 0.5403$, $x_2 = \cos 0.5403 = 0.8576$

MATLAB: `p=(a+b)/2`; else-branch `b=p`; stop when `(b-a)<=tol`.

W9 Newton & secant

$$\textbf{Newton: } x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$\textbf{Secant: } x_{n+1} = x_n - f(x_n)\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}$$

e.g. $f(x) = x^2 - 2$, $f'(x) = 2x$

Newton, $x_0 = 1$:
$x_1 = 1 - \frac{-1}{2} = 1.5$, $x_2 = 1.5 - \frac{0.25}{3} = 1.41667$

Secant, $x_0 = 1$, $x_1 = 2$:
$x_2 = 2 - 2 \cdot \frac{2-1}{2-(-1)} = 1.3333$
$x_3 = 1.3333 - (-0.2222)\frac{1.3333-2}{-0.2222-2} = 1.4$

Derive Newton (3 marks):

1. tangent at x_n: $f(x) \approx f(x_n) + f'(x_n)(x - x_n)$
2. the next guess is where the tangent is 0: $0 = f(x_n) + f'(x_n)(x_{n+1} - x_n)$
3. solve: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

Orders: bisection 1, fixed point 1, secant $\frac{1+\sqrt{5}}{2} \approx 1.618$, Newton 2 (correct digits double). Newton fails if $f'(x_n) = 0$; only linear at a repeated root.

Root-finding formula for $\sqrt[k]{a}$: $f(x) = x^k - a$, apply the method, simplify (use $p^2 - q^2 = (p-q)(p+q)$, $p^3 - q^3 = (p-q)(p^2+pq+q^2)$).

- Newton $\sqrt{a}$: $x_{n+1} = \frac{1}{2}(x_n + \frac{a}{x_n})$; $\sqrt[3]{a}$: $x_{n+1} = \frac{2x_n}{3} + \frac{a}{3x_n^2}$
- Secant $\sqrt{a}$: $x_{n+1} = x_n + \frac{a - x_n^2}{x_n + x_{n-1}}$
- Secant $\sqrt[3]{a}$: $x_{n+1} = x_n - \frac{x_n^3 - a}{x_n^2 + x_n x_{n-1} + x_{n-1}^2}$

MATLAB: `while abs(x-x_old) > tol`; `x - f(x)/fdash(x)`; secant shifts `x0=x1; x1=x2`.

W10 Numerical differentiation

- forward (left end), error $-\frac{h}{2}f''(c)$, $O(h)$:
$$f'(x_0) \approx \frac{f(x_0+h) - f(x_0)}{h}$$
- backward (right end), error $+\frac{h}{2}f''(c)$, $O(h)$:
$$f'(x_0) \approx \frac{f(x_0) - f(x_0-h)}{h}$$
- central (middle), error $-\frac{h^2}{6}f'''(c)$, $O(h^2)$:
$$f'(x_0) \approx \frac{f(x_0+h) - f(x_0-h)}{2h}$$
- second derivative, error $-\frac{h^2}{12}f^{(4)}(c)$, $O(h^2)$:
$$f''(x_0) \approx \frac{f(x_0-h) - 2f(x_0) + f(x_0+h)}{h^2}$$
- 3-point forward, $O(h^2)$:
$$f'(x_0) \approx \frac{-3f(x_0) + 4f(x_0+h) - f(x_0+2h)}{2h}$$

e.g. $f(x) = \ln x$, $x_0 = 2$, $h = 0.1$ (true $f'(2) = 0.5$, $f''(2) = -0.25$)
fwd: $\frac{\ln 2.1 - \ln 2}{0.1} = 0.48790$
bwd: $\frac{\ln 2 - \ln 1.9}{0.1} = 0.51293$
central: $\frac{\ln 2.1 - \ln 1.9}{0.2} = 0.50042$ (best: $O(h^2)$)
$f''(2) \approx \frac{\ln 1.9 - 2\ln 2 + \ln 2.1}{0.01} = -0.25031$

Derive the $f''(x_0)$ formula (4 marks):

1. expand both sides about x_0:
$$f(x_0 \pm h) = f(x_0) \pm hf'(x_0) + \tfrac{h^2}{2}f''(x_0) \pm \tfrac{h^3}{6}f'''(x_0) + \tfrac{h^4}{24}f^{(4)}(c_\pm)$$
2. **add** them (the $\pm$ terms cancel):
$$f(x_0+h) + f(x_0-h) = 2f(x_0) + h^2 f''(x_0) + \tfrac{h^4}{24}\left[f^{(4)}(c_+) + f^{(4)}(c_-)\right]$$
3. rearrange and divide by h^2:
$$f''(x_0) = \frac{f(x_0-h) - 2f(x_0) + f(x_0+h)}{h^2} - \frac{h^2}{12} \cdot \frac{f^{(4)}(c_+) + f^{(4)}(c_-)}{2}$$
4. IVT: the average equals $f^{(4)}(c)$ for some c, so the error is $-\frac{h^2}{12}f^{(4)}(c)$

(For the central $f'(x_0)$: **subtract** instead, divide by $2h$.)

Choice of h (2 marks): too large $\Rightarrow$ **truncation** error; too small $\Rightarrow$ **roundoff** error (cancellation, then $\div$ small h). Best forward $h \approx \sqrt{u} \approx 10^{-8}$.

W11 Integration

$h = \frac{b-a}{n}$, $x_i = a + ih$, n = number of strips ($n+1$ points)

$$T = \tfrac{h}{2}\left[f(x_0) + 2\big(f(x_1) + \cdots + f(x_{n-1})\big) + f(x_n)\right]$$

$$S = \tfrac{h}{3}\left[f(x_0) + 4(\text{odd } i) + 2(\text{even } i) + f(x_n)\right]$$

Errors: T $-\frac{(b-a)h^2}{12}f''(c)$, $O(h^2)$, exact for degree ≤ 1;
S $-\frac{(b-a)h^4}{180}f^{(4)}(c)$, $O(h^4)$, exact for degree ≤ 3. Simpson needs **even** n.

e.g. $\int_1^2 \frac{1}{x}\,dx = \ln 2 = 0.693147$, $n = 4$, $h = 0.25$
f: 1, 0.8, 0.66667, 0.571429, 0.5
$T = \frac{0.25}{2}[1 + 2(0.8 + 0.66667 + 0.571429) + 0.5] = 0.697024$
$S = \frac{0.25}{3}[1 + 4(0.8 + 0.571429) + 2(0.66667) + 0.5] = 0.693254$
Simpson is closer: $O(h^4)$ vs $O(h^2)$

Derive the trapezoidal rule:

1. one strip: $P_1(x) = y_0 + s\Delta y_0$, $x = x_0 + sh$, $dx = h\,ds$
2. integrate ($x_0 \to x_1$ means $s = 0 \to 1$):
$$\int_{x_0}^{x_1} f(x)\,dx \approx \int_0^1 (y_0 + s\Delta y_0)\,h\,ds = h\left(y_0 + \tfrac{1}{2}\Delta y_0\right)$$
3. substitute $\Delta y_0 = y_1 - y_0$: $= \frac{h}{2}(y_0 + y_1)$
4. composite: add the n strips; interior points appear twice

Derive Simpson: same steps with P_2 over two strips ($s = 0 \to 2$):

1. $\int_0^2 \left[y_0 + s\Delta y_0 + \frac{s(s-1)}{2}\Delta^2 y_0\right] h\,ds$
2. $= h\left[2y_0 + 2\Delta y_0 + \frac{1}{3}\Delta^2 y_0\right]$ (since $\int_0^2 \frac{s^2 - s}{2}ds = \frac{1}{3}$)
3. substitute $\Delta y_0 = y_1 - y_0$, $\Delta^2 y_0 = y_2 - 2y_1 + y_0$: $= \frac{h}{3}(y_0 + 4y_1 + y_2)$

MATLAB: `h=(b-a)/n`; points `x(1)..x(n+1)`; `2*f` interior; Simpson `h/3`, even `n`; `x(2)` is x_1 (weight 4).

W12 Triangular systems & LU

- forward sub. (lower), top→down:
$$x_i = \frac{b_i - \sum_{j<i} a_{ij}x_j}{a_{ii}}$$
- backward sub. (upper), bottom→up:
$$x_i = \frac{b_i - \sum_{j>i} a_{ij}x_j}{a_{ii}}$$
- **Explain LU (3 marks):** $A = LU \Rightarrow L(U\mathbf{x}) = \mathbf{b}$. Let $\mathbf{z} = U\mathbf{x}$: solve $L\mathbf{z} = \mathbf{b}$ (forward), then $U\mathbf{x} = \mathbf{z}$ (backward). Reuse L, U for a new $\mathbf{b}$
- **Fill blanks:** a_{ij} = (row i of L)·(col j of U); L has **1s on the diagonal**; pick the equation with one unknown
- a zero or tiny pivot u_{ii} breaks LU $\Rightarrow$ pivoting

e.g. $A = \begin{bmatrix}2&1&1\\4&3&3\\8&7&9\end{bmatrix}$, $\mathbf{b} = \begin{bmatrix}4\\10\\24\end{bmatrix}$
$L = \begin{bmatrix}1&0&0\\2&1&0\\4&3&1\end{bmatrix}$, $U = \begin{bmatrix}2&1&1\\0&1&1\\0&0&2\end{bmatrix}$
one blank: $a_{32} = 7 = \ell_{31}u_{12} + \ell_{32}u_{22} = 4(1) + \ell_{32}(1) \Rightarrow \ell_{32} = 3$
$L\mathbf{z} = \mathbf{b}$: $z_1 = 4$, $z_2 = 10 - 2(4) = 2$, $z_3 = 24 - 4(4) - 3(2) = 2$
$U\mathbf{x} = \mathbf{z}$: $x_3 = \frac{2}{2} = 1$, $x_2 = 2 - 1 = 1$, $x_1 = \frac{4-1-1}{2} = 1$

W13 SPD & Cholesky

- **SPD:** $A^T = A$ and $\mathbf{x}^T A\mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$. 2×2: PD $\iff a > 0$, $ac - b^2 > 0$
- **Cholesky:** $A = LL^T$, a_{ij} = (row i of L)·(row j of L). ! Diagonal ℓ_{jj}: take the **positive** root. $\sqrt{\text{negative}} \Rightarrow$ not SPD
- solve $L\mathbf{z} = \mathbf{b}$ (forward), then $L^T\mathbf{x} = \mathbf{z}$ (**backward**)

Show PD, $A = \begin{bmatrix}a&b\\b&c\end{bmatrix}$ (3 marks):

1. goal: $\mathbf{x}^T A\mathbf{x} > 0$ for every $\mathbf{x} \neq \mathbf{0}$
2. $\mathbf{x}^T A\mathbf{x} = ax_1^2 + 2bx_1x_2 + cx_2^2$
3. complete the square: $= a\left(x_1 + \frac{b}{a}x_2\right)^2 + \left(c - \frac{b^2}{a}\right)x_2^2$
4. both terms ≥ 0, and $= 0$ only if $x_2 = 0$ and $x_1 = 0$, i.e. $\mathbf{x} = \mathbf{0}$

e.g. Cholesky of $A = \begin{bmatrix}4&2&2\\2&5&3\\2&3&6\end{bmatrix}$
$\ell_{11} = \sqrt{4} = 2$, $\ell_{21} = \frac{2}{2} = 1$, $\ell_{31} = \frac{2}{2} = 1$
$\ell_{22} = \sqrt{5 - 1^2} = 2$, $\ell_{32} = \frac{3-(1)(1)}{2} = 1$
$\ell_{33} = \sqrt{6 - 1^2 - 1^2} = 2$ (not -2)
$L = \begin{bmatrix}2&0&0\\1&2&0\\1&1&2\end{bmatrix}$; "Is A SPD?" Yes: symmetric and Cholesky succeeded

Order of accuracy, all methods

- $O(h)$: Euler; forward/backward difference
- $O(h^2)$: Taylor 2, Mod. Euler; central $f'(x_0)$, $f''(x_0)$; trapezoid
- $O(h^4)$: RK4; Simpson
- roots: bisection and fixed point 1; secant 1.618; Newton 2